

LAB 3

Task # 01 (Default Constructor Initialization)

- Create a class named Student with instance variables name and Roll Number.
- Create a default constructor that initializes the name to "Unknown" and Roll Number to 0.
- Create a student object without passing any arguments and print its details.

CODE:

```
class Student {  
  
    String name;  
  
    int rollNumber;  
  
    public Student() {  
  
        name = "Unknown";  
  
        rollNumber = 0;  
  
    }  
  
    public static void main(String[] args) {  
  
        Student student = new Student();  
  
    }  
}
```

```
        System.out.println("Student Name : " + student.name);  
  
        System.out.println("Student Roll-Number : " +  
student.rollNumber);  
  
    }  
}
```

OUTPUT:

```
Student Name : Unknown  
Student Roll-Number : 0
```

Task # 02 (Parameterized Constructor Initialization)

- Add a parameterized constructor to the student class that takes name and roll Number as parameters and initializes the instance variables.
- Create a student object by passing values to the parameterized constructor and print its details.

CODE:

```
class Student {  
  
    String name;  
  
    int rollNumber;  
  
    public Student(String name, int rollNumber) {  
  
        this.name = name;  
  
        this.rollNumber = rollNumber;  
  
    }  
  
    public static void main(String[] args) {  
  
    }  
}
```

```
        Student student = new Student("Muneeb", 27855);  
  
        System.out.println("Student Name : " + student.name);  
  
        System.out.println("Student Roll-Number : " +  
student.rollNumber);  
  
    }  
}
```

OUTPUT:

```
Student Name : Muneeb  
Student Roll-Number : 27855
```

LAB 3

Task # 03 (Method-Based Initialization)

- Add a method setName and setRollNumber to the Student class to set the values of the instance variables.
- Create a Student object using the default constructor.
- Use the setName and setRollNumber methods to initialize the object's values and print the details.

CODE:

```
class Student {  
  
    String name;  
  
    int rollNumber;  
  
    void setName(String name) {  
  
        this.name = name;  
  
    }  
  
    void setRollNumber(int rollNumber) {  
  
        this.rollNumber = rollNumber;  
  
    }  
}
```

```
public static void main(String[] args) {  
  
    Student student = new Student();  
  
    student.setName("Muneeb Mughal");  
  
    student.setRollNumber(27855);  
  
    System.out.println(student.name);  
  
    System.out.println(student.rollNumber);  
  
    }  
}
```

OUTPUT:

```
Muneeb Mughal  
27855
```

LAB 3

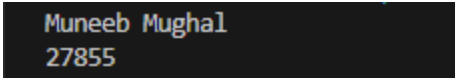
Task # 04 (Combining Constructors and Methods)

- Create a parameterized constructor that initializes only the name and a setRollNumber method to initialize the rollNumber.
- Create a Student object using the parameterized constructor and then use the setRollNumber method to complete the initialization.
- Print the object's details.

CODE:

```
class Student {  
  
    String name;  
  
    int rollNumber;  
  
    Student(String name) {  
  
        this.name = name;  
    }  
  
    void setRollNumber(int rollNumber) {  
  
        this.rollNumber = rollNumber;  
    }  
  
    public static void main(String[] args) {  
  
        Student student = new Student("Muneeb Mughal");  
  
        student.setRollNumber(27855);  
  
        System.out.println(student.name);  
  
        System.out.println(student.rollNumber);  
    }  
}
```

OUTPUT:



```
Muneeb Mughal  
27855
```

LAB 3

Task # 04 (Understanding Reference Variables)

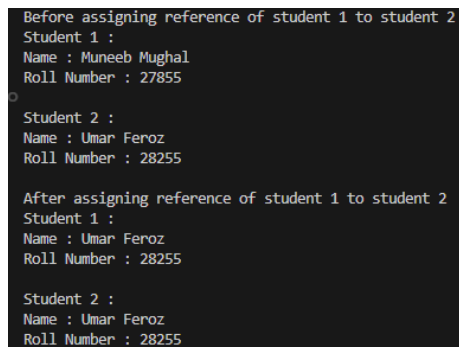
- Create two Student objects, student1 and student2.
- Assign the reference of student1 to student2.
- Modify a property of student1.
- Print the values of both student1 and student2 to observe the effect of reference assignment..

CODE:

```
class Student {  
  
    String name;  
  
    int rollNumber;  
  
    Student(String name, int rollNumber) {  
  
        this.name = name;  
  
        this.rollNumber = rollNumber;  
  
    }  
  
    void displayInfo() {  
  
        System.out.println("Name : " + this.name);  
  
        System.out.println("Roll Number : " + this.rollNumber + "\n");  
  
    }  
  
    public static void main(String[] args) {  
  
        Student student1 = new Student("Muneeb Mughal", 27855);  
  
        Student student2 = new Student("Umar Feroz", 28255);  
  
        System.out.println("Before assigning reference of student 1 to  
student 2");  
  
        System.out.println("Student 1 :");  
  
        student1.displayInfo()
```

```
System.out.println("Student 2 :");  
  
        student2.displayInfo();  
  
        student1 = student2;  
  
        System.out.println("After assigning reference of student 1  
to student 2");  
  
        System.out.println("Student 1 :");  
  
        student1.displayInfo();  
  
        System.out.println("Student 2 :");  
  
        student2.displayInfo();  
  
    }  
}
```

OUTPUT:



```
Before assigning reference of student 1 to student 2  
Student 1 :  
Name : Muneeb Mughal  
Roll Number : 27855  
  
Student 2 :  
Name : Umar Feroz  
Roll Number : 28255  
  
After assigning reference of student 1 to student 2  
Student 1 :  
Name : Umar Feroz  
Roll Number : 28255  
  
Student 2 :  
Name : Umar Feroz  
Roll Number : 28255
```